Stochastic Calculus, Courant Institute, Fall 2020
http://www.math.nyu.edu/faculty/goodman/teaching/StochCalc2020/index.html

Week 6

Jonathan Goodman, November, 2020

Tentative (more text, no more exercises)

The two topics for this week both involve putting integrals in the exponential. For the *Feynman Kac formula*, the integral is dt. For the *Girsanov change of measure formula*, the integral is dW_t. If X_t is an Ito process, then another Ito process is

$$Y_t = e^{X_t} \; .$$

We can apply Ito's lemma with $f(x) = e^x$, and $\partial_x f(x) = e^x = f(x)$, and $\partial_x^2 f(x) = f(x)$. The result is, using $Y_t = e^{X_t}$ on the right side

$$dY_t = Y_t \, dX_t + \frac{1}{2} Y_t \, (dX_t)^2 \; . \tag{1}$$

If X_t is a dt integral (Feynman Kac), then the second term on the right is zero.

The Feynman Kac "formula" is used in finance to study models of fluctuating interest rates. Suppose the interest rate at time t is X_t, and a "money market account" at time t has value M_t. Then $M_{t+dt} = (1+X_t dt)M_t$. This is expressed as

$$M_T = M_0 e^{\int_0^T X_t dt} \; .$$

The Feynman Kac formula is related to expected values of expressions like this.

1 Feynman Kac Formula

The *Feynman Kac* formula, as the term is used in finance, is a relationship between a multiplicative value function and the backward equation it satisfies. Suppose X_t is a diffusion process that satisfies the SDE

$$dX_t = a(X_t) \, dt + b(X_t) \, dW_t \; . \tag{2}$$

A multiplicative functional is a function of the diffusion process path of the form

$$e^{\int_0^T V(X_s) \, ds} \; . \tag{3}$$

This is called *multiplicative* because you can think of it as multiplying together the individual pieces $V(X_s)ds$. You can see this more explicitly by choosing a finite Δt approximation to the integral. As usual, $t_k = k\Delta t$. We can approximate the functional as

$$e^{\int_0^T V(X_s)\,ds} \approx \exp\left\{\sum_{t_k<T} V(X_{t_k})\Delta t\right\}$$
$$= \prod_{t_k<T} e^{V(X_{t_k})\Delta t}\;. \tag{4}$$

Moreover, you can use the exponential approximation $e^\epsilon \approx 1+\epsilon$ to write the approximation is

$$\prod_{t_k<T} e^{V(X_{t_k})\Delta t} \approx \prod_{t_k<T} \left[\,1 + V(X_{t_k})\,\Delta t\,\right]\;.$$

To be even more explicit, suppose you define the product of the first n factors to be

$$P_n = \prod_{k=0}^{n-1} \left[\,1 + V(X_{t_k})\,\Delta t\,\right]\;.$$

Then

$$P_{n+1} = \left[\,1 + V(X_{t_n})\,\Delta t\,\right] P_n\;.$$

This has the interpretation as an interest payment. The "account" at time n is P_n. As we go from time n to time $n+1$, we receive a payment of $V_n \Delta t P_n$. This is the payment, for example, if V is an interest rate in years and Δt is measured as a fraction of a year. The final amount P_n is random because the interest rates $V_k = V(X_{t_k})$ are random.

The value function corresponding to the multiplicative functional is

$$f(x,t) = \mathrm{E}\left[\, e^{\int_t^T V(X_s)\,ds} \mid X_t = t\,\right]\;. \tag{5}$$

There may be other ways to define a value function if we are interested in the functional with the integral starting at time $t=0$. If you take the integral starting at time t, then functional in the value function (5) does not depend on X_s for $s<t$. That means that the expectation on the right is a function of x and t alone, not the path that got you there.

We can find the backward equation PDE that f satisfies using a variant of the martingale method we used in Week 4. Consider a small time dt, take d of both sides, use Ito's lemma, and set the dt parts equal, ignoring the dW part. On the left, you have

$$df(X_t,t) = \partial_t f dt + \partial_x f\, dX + \frac{1}{2}\partial_x^2 f (dX)^2\;.$$

We identify the infinitesimal mean (the dt part) from the SDE (2) using the Ito rule $(dX)^2 = b^2 dt$. The result is

$$df(X_t,t) = \partial_t f dt + a(x)\partial_x f dt + \frac{1}{2}b^2(x)\partial_x^2 f + \partial_x f b(x) dW_t\;.$$

On the right, we can use the Ito calclation (1) to get

$$
\begin{aligned}
de^{\int_t^T V(X_s)\,ds} &= d\left(\int_t^T V(X_s)ds\right) e^{\int_t^T V(X_s)\,ds} \\
&= -V(x)\,dt\,e^{\int_t^T V(X_s)\,ds} \; .
\end{aligned}
$$

For the second line, note that if $V > 0$, the integral gets smaller as t increases, and that we are conditioning on $X_t = x$. We put this into the right side of (5), combine terms, drop the common dt factor, and get

$$
\partial_t f(x,t) + a(x)\partial_x f(x,t) + \frac{1}{2} b^2(x)\partial_x^2 f(x,t) + V(x) f(x,t) = 0 \; . \tag{6}
$$

This is the backward equation for the multiplicative functional. The final condition "obviously" (think about it) is $V(x,T) = 1$.

The examples are all rather complicated, unfortunately. Take X_t to be Brownian motion and $V(x) = x$. Then the backward equation is

$$
\partial_t f + \frac{1}{2}\partial_x^2 f + xf = 0 \; . \tag{7}
$$

You can calculate the solution explicitly as in Exercise 2. Or you can guess by trial and error. Either way, we come to the ansatz

$$
f(x,t) = e^{A(t)x + B(t)} \; .
$$

We plug this into the backward equation (7) using the calculations

$$
\begin{aligned}
\partial_t f &= \left(\dot{A}x + \dot{B}\right) f \\
\partial_x^2 f &= \partial_x \left[A e^{Ax+B}\right] = A^2 f \\
xf &= xf \; .
\end{aligned}
$$

This gives

$$
\dot{A}xf + \dot{B}f + \frac{1}{2}A^2 f + xf = 0 \; .
$$

This works if we set the coefficient of x to zero, and the constant term (as a function of x) to zero. This gives the equations

$$
\begin{aligned}
\dot{A} + 1 &= 0 \\
\dot{B} + \frac{1}{2}A^2 &= 0 \; .
\end{aligned}
$$

The first equation, together with the final condition $A(T) = 0$ (why?)) gives $A(t) = T - t$. The second equation becomes

$$
\dot{B} = -\frac{1}{2}(T-t)^2 \; .
$$

The solution is $B(t) = \frac{1}{6}(T-t)^3$. The full solution is

$$f(x,t) = e^{\frac{1}{6}(T-t)^3} e^{(T-t)x} \ .$$

The history of this formula starts in the late 1940's when physicist Richard Feynman wrote a formula for the solution of the Schrödinger equation as a "path integral", which was a kind of infinite product of infinitesimal pieces something like (4). Mathematicians reacted badly to this "Feynman integral" because it does not correspond to an integral in the sense of measure theory. About 15 years later, mathematician Mark Kac found that a similar formula gave an expression for the solution of the backward equation (6). The part of Feynman's formula that mathematicians objected to turned into "Wiener measure", which is the probability distribution of Brownian motion paths. This had the advantage of being mathematically rigorous. By the way, you pronounce "Kac" as "cats". His name is Polish. People who immigrated from Poland before him spelled their names as "Katz". For example, there is a "Katz Delicatessen" on Houston Street not far from NYU.

2 Interest rate models

The financial services industry (as opposed to pure traders) is heavily focused on loans and financial instruments that depend on interest rates. There is a range of stochastic models of interest rates, but the simple models involve only the *short rate* (also called *overnight rate*), which is the interest rate (in %/year, say) that is paid on a loan where you "know" it will be repaid very soon (e.g., the next day). Let us call this rate R_t. One model is

$$dR_t = -\gamma(R_t - r_0)\, dt + \sigma dW_t \ . \tag{8}$$

This is a mean-reverting process centered on an average long term average rate r_0.

A *floating rate* loan is a loan so that the interest rate at time t is related to an index of short rate loans at that time such as LIBOR (google this). We could use the model (8) as a model of this fluctuating rate. Suppose you make a loan with floating interest rate R_t to be repaid at time T. The total repayment amount is a multiplicative function of the initial amount

$$M_T = M_0 e^{\int_0^T R_s\, ds} \ .$$

We may as well set $M_0 = 1$ for simplicity. The value function satisfies the backward equation is

$$\partial_t f - \gamma(r - r_0)\partial_r f + \frac{\sigma^2}{2}\partial_r^2 f - rf = 0 \ .$$

This has an ansatz solution of the form

$$f(r,t) = e^{A(t)r + B(t)} \ .$$

Try it and see. Models like this are called *affine* because the exponent is an affine function of r.